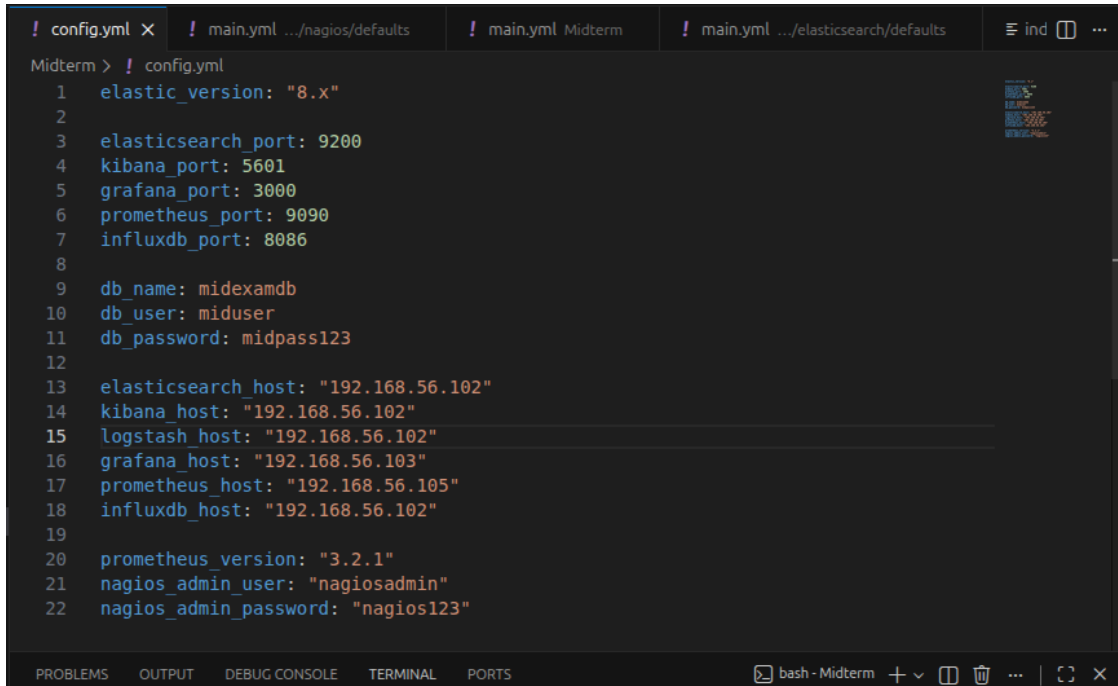


Name: Fetalino Jayvee M.	Date Performed: 12/03/2026
Course/Section: CPE212/CPE31S1	Date Submitted: 12/03/2026
Instructor: Engr. Robin Valenzuela	Semester and SY:
Midterm Skills Exam: Install, Configure, and Manage Log Monitoring tools	
1. Objectives	
Create and design a workflow that installs, configure and manage enterprise availability, performance and log monitoring tools using Ansible as an Infrastructure as Code (IaC) tool.	
2. Instructions	
1. Create a repository in your GitHub account and label it CPE_MIDEXAM_SURNAME. 2. Clone the repository and do the following: 2.1. Create an Ansible playbook that does the following with an input of a config.yaml file and arranged Inventory file: 2.2. Install and configure Elastic Stack in separate hosts (Elastic Search, Kibana, Logstash) • Install Nagios in one host 2.3. Install Grafana,Prometheus and Influxdb in seperate hosts (Influxdb,Grafana,Prometheus) 2.4. Install Lamp Stack in separate hosts (Httpd + Php,Mariadb) 3. Document all your tasks using this document. Provide proofs of all the ansible playbooks codes and successful installations. 4. Document the push and commit from the local repository to GitHub. 5. Finally, paste also the link of your GitHub repository in the documentation.	

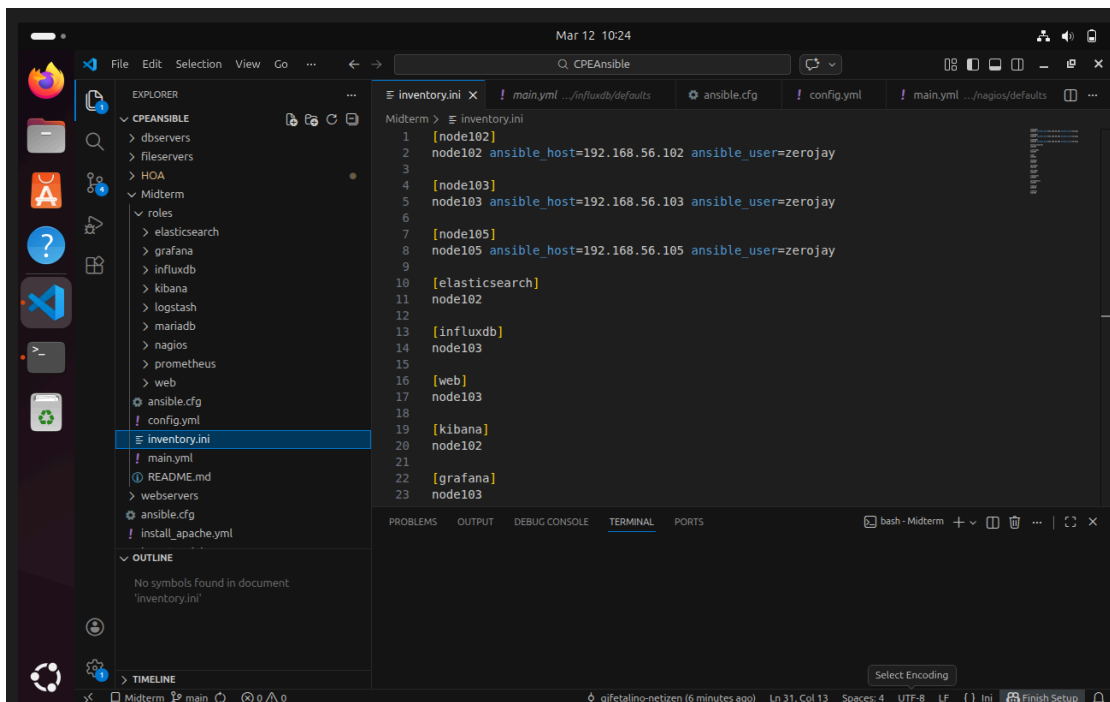
3. Output (screenshots and explanations)

First I created a config.yml file so I can use it later for vars in the main.yml also for configuring datas needed for the installation



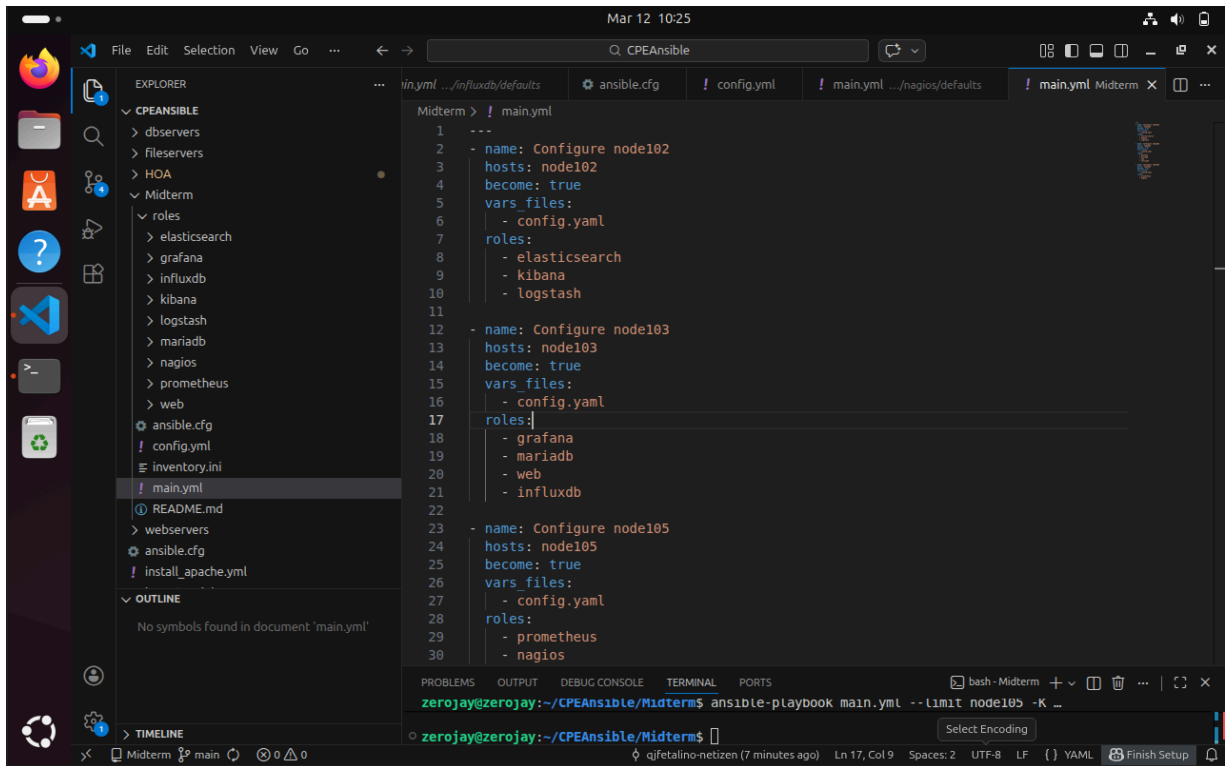
```
! config.yml x ! main.yml .../nagios/defaults ! main.yml Midterm ! main.yml .../elasticsearch/defaults ind ...
Midterm > ! config.yml
1 elastic_version: "8.x"
2
3 elasticsearch_port: 9200
4 kibana_port: 5601
5 grafana_port: 3000
6 prometheus_port: 9090
7 influxdb_port: 8086
8
9 db_name: midexamdb
10 db_user: miduser
11 db_password: midpass123
12
13 elasticsearch_host: "192.168.56.102"
14 kibana_host: "192.168.56.102"
15 logstash_host: "192.168.56.102"
16 grafana_host: "192.168.56.103"
17 prometheus_host: "192.168.56.105"
18 influxdb_host: "192.168.56.102"
19
20 prometheus_version: "3.2.1"
21 nagios_admin_user: "nagiosadmin"
22 nagios_admin_password: "nagios123"
```

Next is configuring my inventory.ini which the ip of my hosts are stored



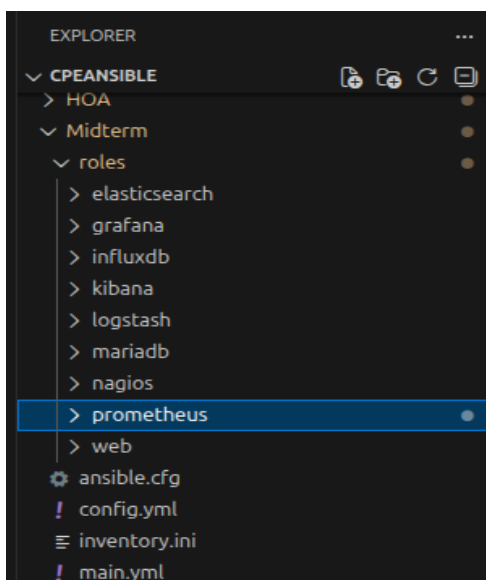
```
Mar 12 10:24
CPEAnsible
File Edit Selection View Go ...
EXPLORER
CPEANSIBLE
> dbservers
> fileservers
> HOA
> Midterm
  > roles
    > elasticsearch
    > grafana
    > influxdb
    > kibana
    > logstash
    > mariadb
    > nagios
    > prometheus
    > web
  > ansible.cfg
  ! config.yml
  inventory.ini
  ! main.yml
  README.md
  > webservers
  > ansible.cfg
  ! install_apache.yml
OUTLINE
No symbols found in document
'inventory.ini'
TIMELINE
Midterm main
inventory.ini
1 [node102]
2 node102 ansible_host=192.168.56.102 ansible_user=zerojay
3
4 [node103]
5 node103 ansible_host=192.168.56.103 ansible_user=zerojay
6
7 [node105]
8 node105 ansible_host=192.168.56.105 ansible_user=zerojay
9
10 [elasticsearch]
11 node102
12
13 [influxdb]
14 node103
15
16 [web]
17 node103
18
19 [kibana]
20 node102
21
22 [grafana]
23 node103
PROBLEMS OUTPUT DEBUG CONSOLE TERMINAL PORTS
bash - Midterm
```

Next is my main.yml where I'll call the roles for its installation in respected nodes or host.



```
1 ---
2 - name: Configure node102
3   hosts: node102
4   become: true
5   vars_files:
6     - config.yaml
7   roles:
8     - elasticsearch
9     - kibana
10    - logstash
11
12 - name: Configure node103
13   hosts: node103
14   become: true
15   vars_files:
16     - config.yaml
17   roles:
18     - grafana
19     - mariadb
20     - web
21     - influxdb
22
23 - name: Configure node105
24   hosts: node105
25   become: true
26   vars_files:
27     - config.yaml
28   roles:
29     - prometheus
30     - nagios
```

Then I'll configure each and every roles that I created using ansible-galaxy and searching the web for the installation guide for each of the application



Now when I run the main.yml all of the file succeeded

The screenshot displays the VS Code interface during the setup of Prometheus. The Explorer pane on the left shows the project structure with files like `ansible.cfg`, `config.yml`, `main.yml`, and `roles`. The main editor shows the content of `main.yml`, which defines the Prometheus service with version 3.2.1, port 9090, and host 192.168.56.105. The Output pane at the bottom shows the execution of the playbook, including a pre-flight check and the status of tasks like 'Enable Apache' and 'Enable Nagios'.

```
elasticsearch main task.yml
```

The screenshot shows the VS Code editor with the 'main.yml' file open. The file content is as follows:

```

1 ---
2 - name: Install Elasticsearch dependencies
3   apt:
4     name:
5       - apt-transport-https
6       - wget
7       - gnupg
8       - openjdk-17-jdk
9     state: present
10    update_cache: yes
11
12 - name: Add Elastic GPG key
13   shell: |
14     wget -qO - https://artifacts.elastic.co/GPG-KEY-elasticsearch | gpg --dearmor
15     args:
16       creates: /usr/share/keyrings/elastic-keyring.gpg
17
18 - name: Add Elastic APT repository
19   apt_repository:
20     repo: "deb [signed-by=/usr/share/keyrings/elastic-keyring.gpg] https://artifac
21     state: present
22     filename: elastic
23
24 - name: Install Elasticsearch
  
```

The terminal output shows the execution of the playbook:

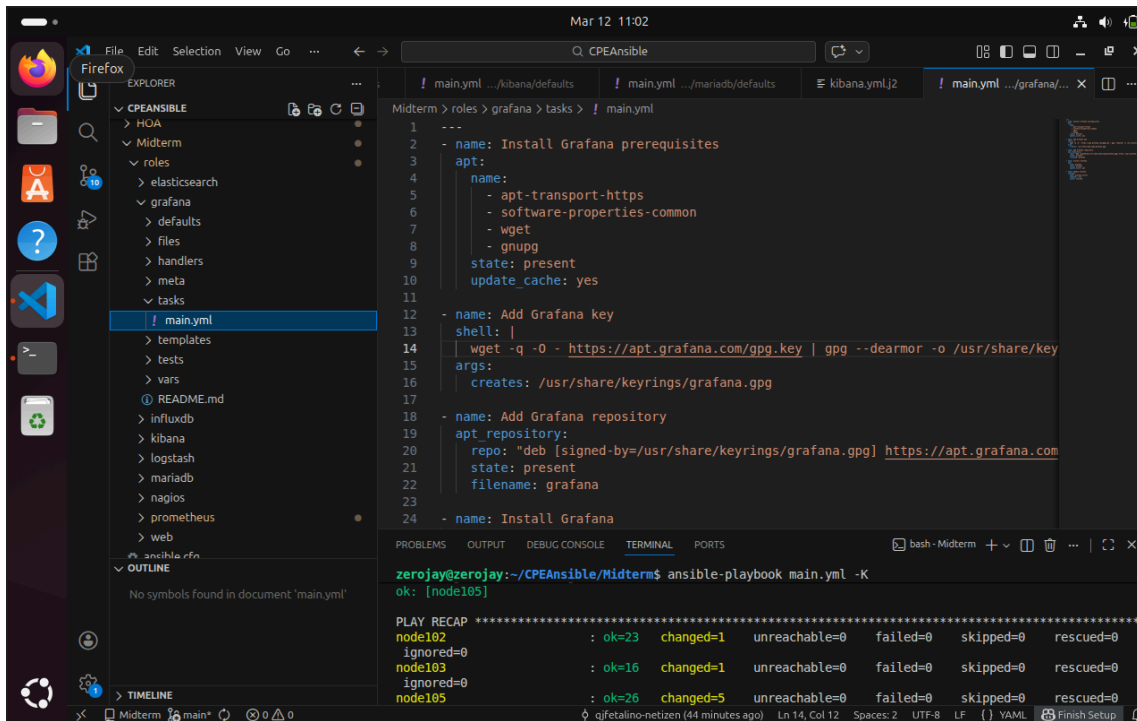
```

zeroday@zeroday:~/CPEAnsible/Midterm$ ansible-playbook main.yml -K
TASK [nagios : Enable Apache] *****
ok: [node105]

TASK [nagios : Enable Nagios] *****
ok: [node105]

PLAY RECAP *****
node102                : ok=23  changed=1    unreachable=0    failed=0    skipped=0    rescued=0
  
```

Grafana main task.yml



The screenshot shows the VS Code editor with the file explorer on the left displaying the project structure. The main editor shows the `main.yml` file under the `grafana` role. The terminal at the bottom shows the execution of the Ansible playbook.

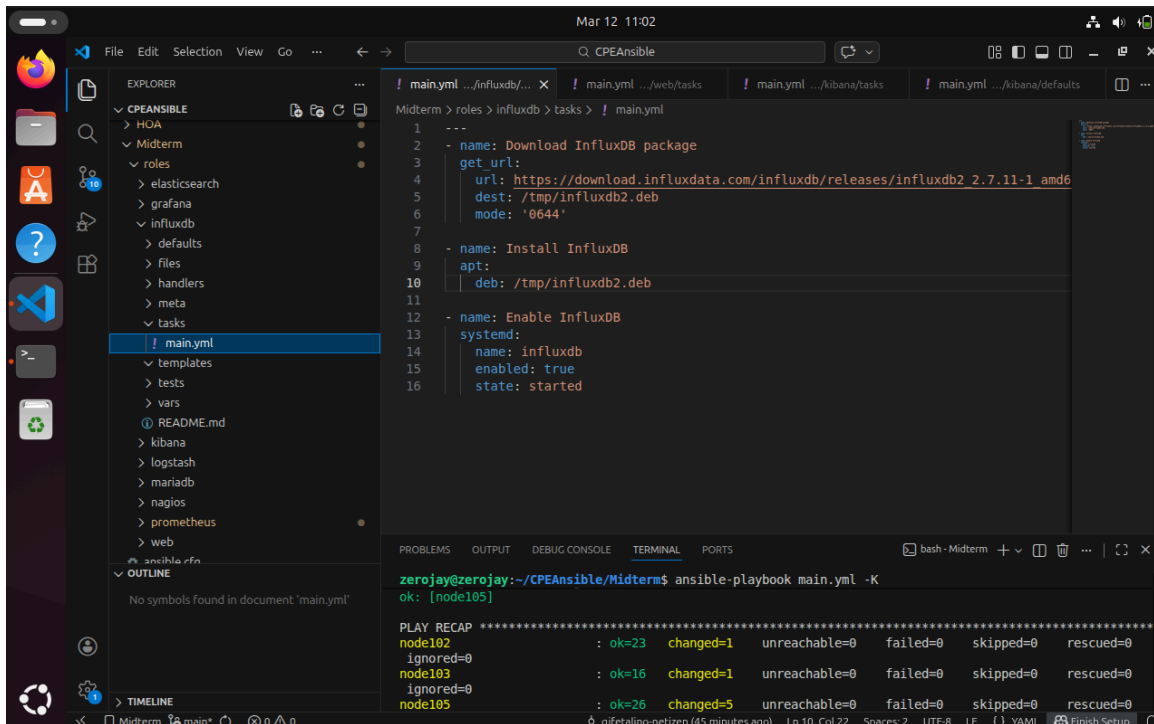
```
1 ---
2 - name: Install Grafana prerequisites
3   apt:
4     name:
5       - apt-transport-https
6       - software-properties-common
7       - wget
8       - gnupg
9     state: present
10    update_cache: yes
11
12 - name: Add Grafana key
13   shell: |
14     wget -q -O - https://apt.grafana.com/gpg.key | gpg --dearmor -o /usr/share/key
15   args:
16     creates: /usr/share/keyrings/grafana.gpg
17
18 - name: Add Grafana repository
19   apt_repository:
20     repo: "deb [signed-by=/usr/share/keyrings/grafana.gpg] https://apt.grafana.com
21     state: present
22     filename: grafana
23
24 - name: Install Grafana
```

Terminal output:

```
zerojay@zerojay:~/CPEAnsible/Midterm$ ansible-playbook main.yml -K
ok: [node105]

PLAY RECAP *****
node102      : ok=23  changed=1  unreachable=0  failed=0  skipped=0  rescued=0
ignored=0
node103      : ok=16  changed=1  unreachable=0  failed=0  skipped=0  rescued=0
ignored=0
node105      : ok=26  changed=5  unreachable=0  failed=0  skipped=0  rescued=0
ignored=0
```

Influxdb main task.yml



The screenshot shows the VS Code editor with the file explorer on the left displaying the project structure. The main editor shows the `main.yml` file under the `influxdb` role. The terminal at the bottom shows the execution of the Ansible playbook.

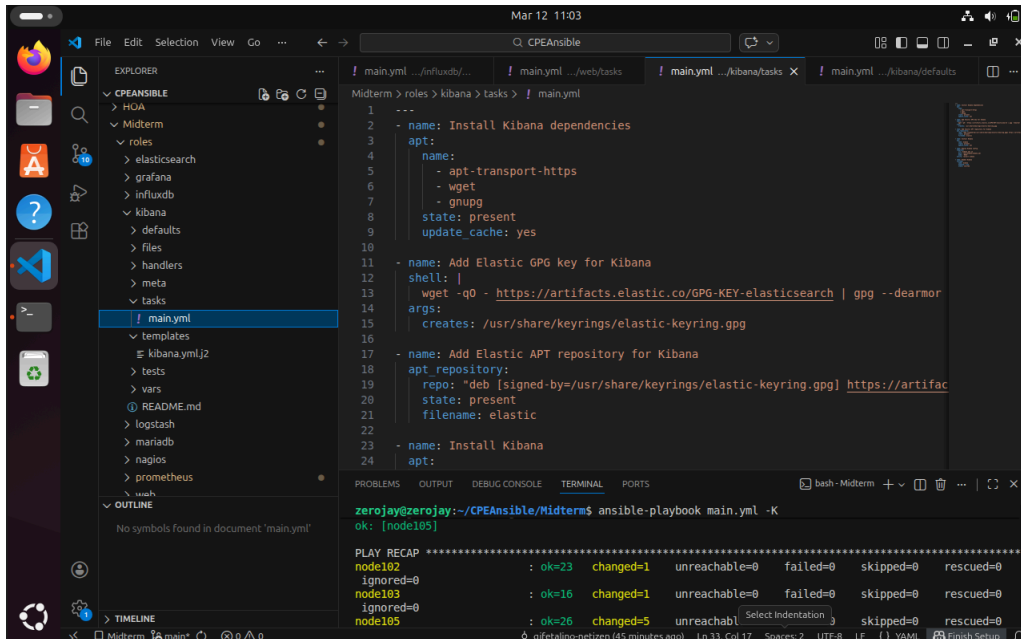
```
1 ---
2 - name: Download InfluxDB package
3   get_url:
4     url: https://download.influxdata.com/influxdb/releases/influxdb2_2.7.11-1 amd6
5     dest: /tmp/influxdb2.deb
6     mode: '0644'
7
8 - name: Install InfluxDB
9   apt:
10    deb: /tmp/influxdb2.deb
11
12 - name: Enable InfluxDB
13   systemd:
14     name: influxdb
15     enabled: true
16     state: started
```

Terminal output:

```
zerojay@zerojay:~/CPEAnsible/Midterm$ ansible-playbook main.yml -K
ok: [node105]

PLAY RECAP *****
node102      : ok=23  changed=1  unreachable=0  failed=0  skipped=0  rescued=0
ignored=0
node103      : ok=16  changed=1  unreachable=0  failed=0  skipped=0  rescued=0
ignored=0
node105      : ok=26  changed=5  unreachable=0  failed=0  skipped=0  rescued=0
ignored=0
```

kibana main task.yml



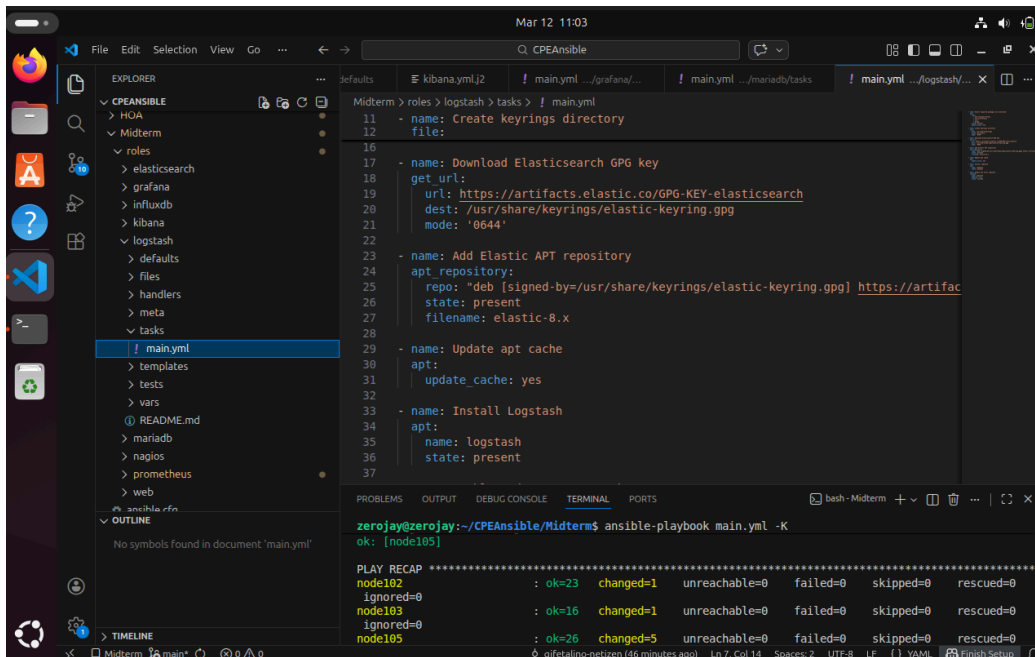
The screenshot shows the VS Code editor with the file explorer on the left displaying the project structure. The main editor shows the content of `main.yml` under the `roles > kibana > tasks` path. The terminal at the bottom shows the execution of the Ansible playbook.

```
1 ---
2 - name: Install Kibana dependencies
3   apt:
4     name:
5       - apt-transport-https
6       - wget
7       - gnupg
8     state: present
9     update_cache: yes
10
11 - name: Add Elastic GPG key for Kibana
12   shell: |
13     wget -qO - https://artifacts.elastic.co/GPG-KEY-elasticsearch | gpg --dearmor
14     args:
15       creates: /usr/share/keyrings/elastic-keyring.gpg
16
17 - name: Add Elastic APT repository for Kibana
18   apt_repository:
19     repo: "deb [signed-by=/usr/share/keyrings/elastic-keyring.gpg] https://artifac
20     state: present
21     filename: elastic
22
23 - name: Install Kibana
24   apt:
```

```
zerojay@zerojay:~/CPEAnsible/Midterm$ ansible-playbook main.yml -K
ok: [node105]

PLAY RECAP *****
node102                : ok=23  changed=1  unreachable=0  failed=0  skipped=0  rescued=0
node103                : ok=16  changed=1  unreachable=0  failed=0  skipped=0  rescued=0
node105                : ok=26  changed=5  unreachable=0  failed=0  skipped=0  rescued=0
```

logstash main task.yml



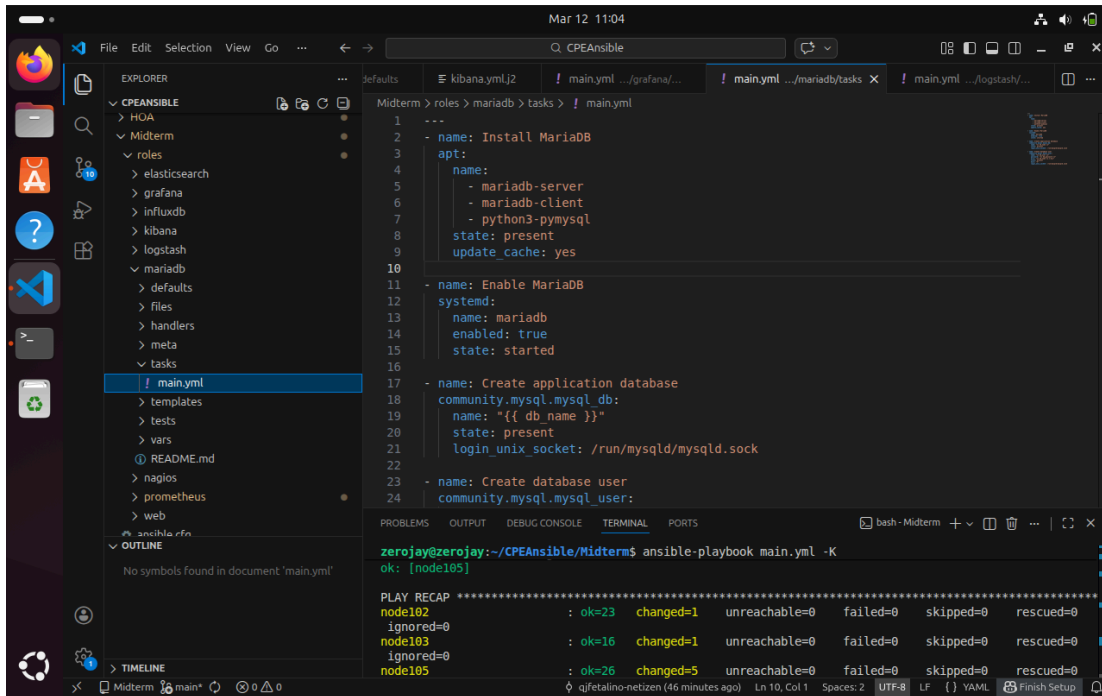
The screenshot shows the VS Code editor with the file explorer on the left displaying the project structure. The main editor shows the content of `main.yml` under the `roles > logstash > tasks` path. The terminal at the bottom shows the execution of the Ansible playbook.

```
11 - name: Create keyrings directory
12   file:
13     path: /usr/share/keyrings
14     state: directory
15
16 - name: Download Elasticsearch GPG key
17   get_url:
18     url: https://artifacts.elastic.co/GPG-KEY-elasticsearch
19     dest: /usr/share/keyrings/elastic-keyring.gpg
20     mode: '0644'
21
22 - name: Add Elastic APT repository
23   apt_repository:
24     repo: "deb [signed-by=/usr/share/keyrings/elastic-keyring.gpg] https://artifac
25     state: present
26     filename: elastic-8.x
27
28 - name: Update apt cache
29   apt:
30     update_cache: yes
31
32 - name: Install Logstash
33   apt:
34     name: logstash
35     state: present
36
37
```

```
zerojay@zerojay:~/CPEAnsible/Midterm$ ansible-playbook main.yml -K
ok: [node105]

PLAY RECAP *****
node102                : ok=23  changed=1  unreachable=0  failed=0  skipped=0  rescued=0
node103                : ok=16  changed=1  unreachable=0  failed=0  skipped=0  rescued=0
node105                : ok=26  changed=5  unreachable=0  failed=0  skipped=0  rescued=0
```

maria db main task.yml



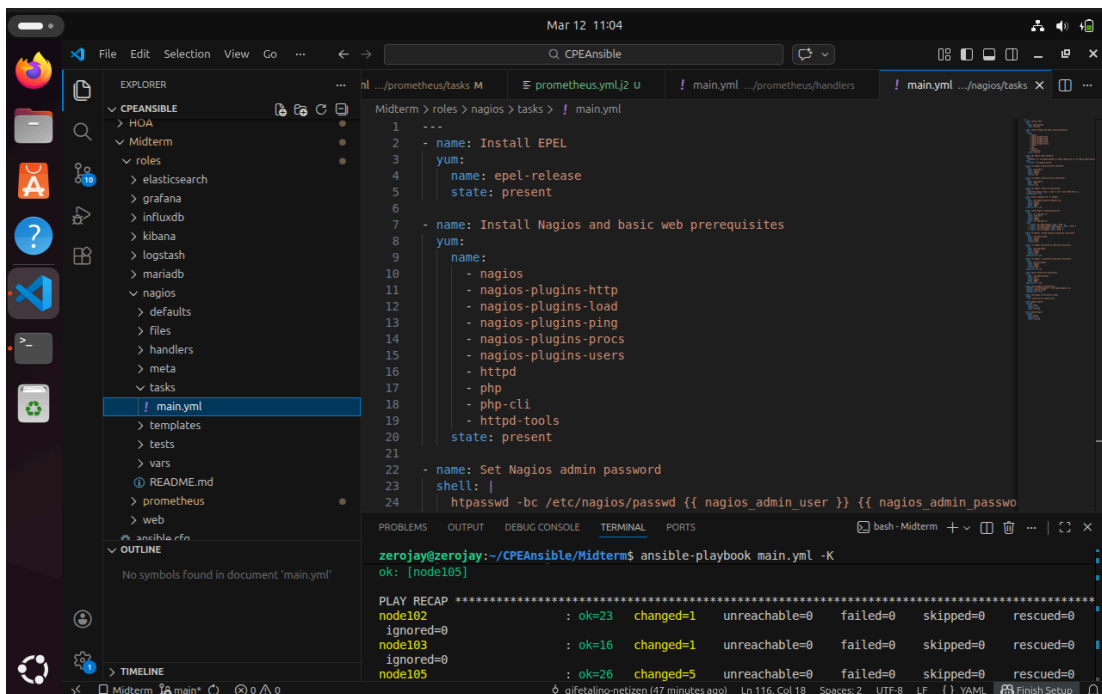
The screenshot shows the Visual Studio Code editor with the file explorer on the left displaying the project structure. The main editor window shows the content of 'main.yml' under the path 'Midterm > roles > mariadb > tasks > ! main.yml'. The file contains four tasks: 'Install MariaDB', 'Enable MariaDB', 'Create application database', and 'Create database user'. The terminal at the bottom shows the command 'ansible-playbook main.yml -K' being executed, resulting in a successful run across three nodes (node102, node103, node105) with a recap of the changes made.

```
1 ---
2 - name: Install MariaDB
3   apt:
4     name:
5       - mariadb-server
6       - mariadb-client
7       - python3-pymysql
8     state: present
9     update_cache: yes
10
11 - name: Enable MariaDB
12   systemd:
13     name: mariadb
14     enabled: true
15     state: started
16
17 - name: Create application database
18   community.mysql.mysql_db:
19     name: "{{ db_name }}"
20     state: present
21     login_unix_socket: /run/mysqld/mysqld.sock
22
23 - name: Create database user
24   community.mysql.mysql_user:
```

zerojay@zerojay:~/CPEAnsible/Midterm\$ ansible-playbook main.yml -K
ok: [node105]

PLAY RECAP *****
node102 : ok=23 changed=1 unreachable=0 failed=0 skipped=0 rescued=0
ignored=0
node103 : ok=16 changed=1 unreachable=0 failed=0 skipped=0 rescued=0
ignored=0
node105 : ok=26 changed=5 unreachable=0 failed=0 skipped=0 rescued=0
ignored=0

nagios main task.yml



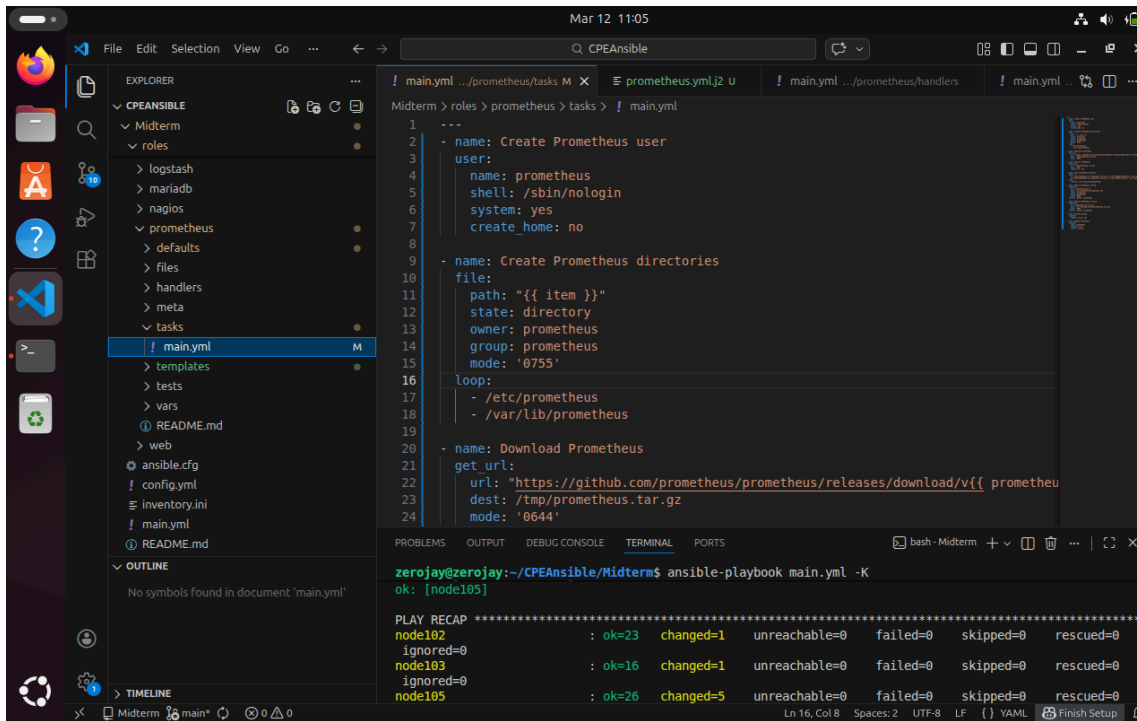
The screenshot shows the Visual Studio Code editor with the file explorer on the left displaying the project structure. The main editor window shows the content of 'main.yml' under the path 'Midterm > roles > nagios > tasks > ! main.yml'. The file contains three tasks: 'Install EPEL', 'Install Nagios and basic web prerequisites', and 'Set Nagios admin password'. The terminal at the bottom shows the command 'ansible-playbook main.yml -K' being executed, resulting in a successful run across three nodes (node102, node103, node105) with a recap of the changes made.

```
1 ---
2 - name: Install EPEL
3   yum:
4     name: epel-release
5     state: present
6
7 - name: Install Nagios and basic web prerequisites
8   yum:
9     name:
10       - nagios
11       - nagios-plugins-http
12       - nagios-plugins-load
13       - nagios-plugins-ping
14       - nagios-plugins-procs
15       - nagios-plugins-users
16       - httpd
17       - php
18       - php-cli
19       - httpd-tools
20     state: present
21
22 - name: Set Nagios admin password
23   shell: |
24     htpasswd -bc /etc/nagios/passwd {{ nagios_admin_user }} {{ nagios_admin_passwo
```

zerojay@zerojay:~/CPEAnsible/Midterm\$ ansible-playbook main.yml -K
ok: [node105]

PLAY RECAP *****
node102 : ok=23 changed=1 unreachable=0 failed=0 skipped=0 rescued=0
ignored=0
node103 : ok=16 changed=1 unreachable=0 failed=0 skipped=0 rescued=0
ignored=0
node105 : ok=26 changed=5 unreachable=0 failed=0 skipped=0 rescued=0
ignored=0

prometheus



The screenshot shows the VS Code interface with the 'main.yml' file open in the editor. The file contains an Ansible playbook for Prometheus. The terminal output shows the execution of the playbook, which successfully creates the Prometheus user and directories on all three nodes (node102, node103, node105).

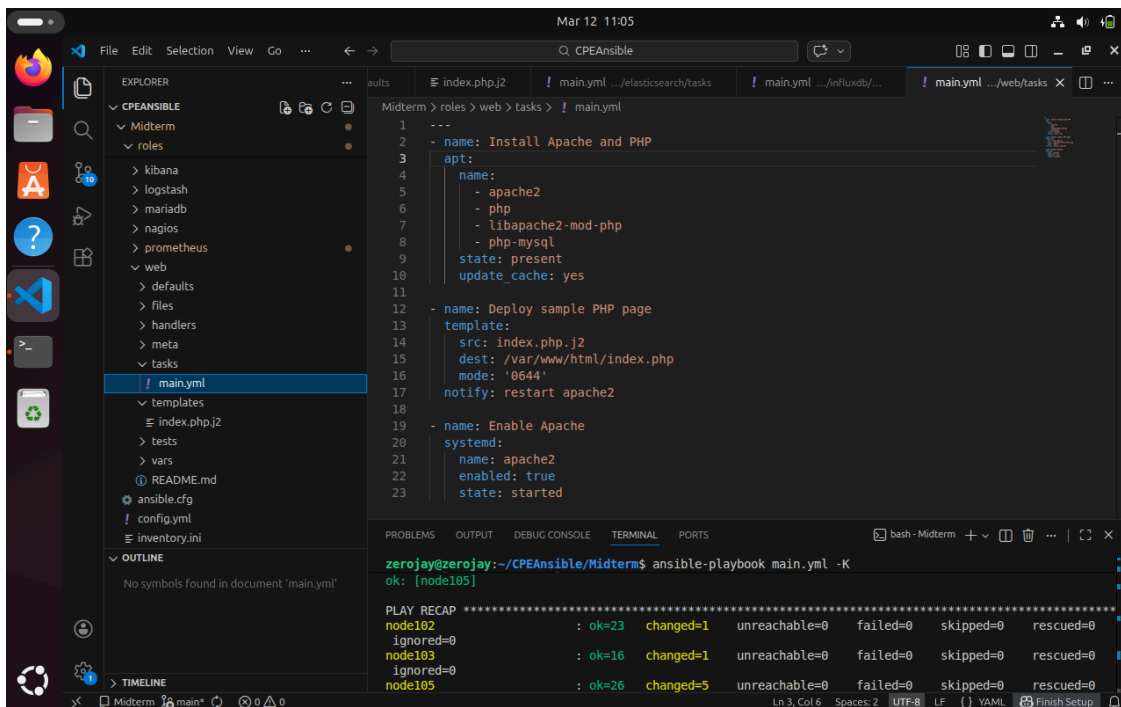
```
1 ---
2 - name: Create Prometheus user
3   user:
4     name: prometheus
5     shell: /sbin/nologin
6     system: yes
7     create_home: no
8
9 - name: Create Prometheus directories
10  file:
11    path: "{{ item }}"
12    state: directory
13    owner: prometheus
14    group: prometheus
15    mode: '0755'
16  loop:
17    - /etc/prometheus
18    - /var/lib/prometheus
19
20 - name: Download Prometheus
21  get_url:
22    url: "https://github.com/prometheus/prometheus/releases/download/v{{ prometheus_version }}"
23    dest: /tmp/prometheus.tar.gz
24    mode: '0644'
```

Terminal Output:

```
zerojay@zerojay:~/CPEAnsible/Midterm$ ansible-playbook main.yml -K
ok: [node105]

PLAY RECAP *****
node102                : ok=23  changed=1  unreachable=0  failed=0  skipped=0  rescued=0
ignored=0
node103                : ok=16  changed=1  unreachable=0  failed=0  skipped=0  rescued=0
ignored=0
node105                : ok=26  changed=5  unreachable=0  failed=0  skipped=0  rescued=0
ignored=0
```

web apache



The screenshot shows the VS Code interface with the 'main.yml' file open in the editor. The file contains an Ansible playbook for Apache. The terminal output shows the execution of the playbook, which successfully installs Apache and PHP on all three nodes (node102, node103, node105).

```
1 ---
2 - name: Install Apache and PHP
3   apt:
4     name:
5       - apache2
6       - php
7       - libapache2-mod-php
8       - php-mysql
9     state: present
10    update_cache: yes
11
12 - name: Deploy sample PHP page
13  template:
14    src: index.php.j2
15    dest: /var/www/html/index.php
16    mode: '0644'
17  notify: restart apache2
18
19 - name: Enable Apache
20  systemd:
21    name: apache2
22    enabled: true
23    state: started
```

Terminal Output:

```
zerojay@zerojay:~/CPEAnsible/Midterm$ ansible-playbook main.yml -K
ok: [node105]

PLAY RECAP *****
node102                : ok=23  changed=1  unreachable=0  failed=0  skipped=0  rescued=0
ignored=0
node103                : ok=16  changed=1  unreachable=0  failed=0  skipped=0  rescued=0
ignored=0
node105                : ok=26  changed=5  unreachable=0  failed=0  skipped=0  rescued=0
ignored=0
```


Centos Nagios checkout

```
zerojay@localhost:~ - systemctl status nagios

● nagios.service - Nagios Core 4.5.9
   Loaded: loaded (/usr/lib/systemd/system/nagios.service; enabled; preset: disabled)
   Active: active (running) since Thu 2026-03-12 18:10:21 PST; 56min ago
   Invocation: f9b4bc983f8a4bb7ac1c097c7ebd23a7
     Docs: https://www.nagios.org/documentation
   Main PID: 1364 (nagios)
    Tasks: 11 (limit: 34941)
   Memory: 56.9M (peak: 59.8M)
      CPU: 5.986s
   CGroup: /system.slice/nagios.service
           └─1364 /usr/sbin/nagios -d /etc/nagios/nagios.cfg
             └─1369 /usr/sbin/nagios --worker /var/spool/nagios/cmd/nagios.qh
               └─1371 /usr/sbin/nagios --worker /var/spool/nagios/cmd/nagios.qh
                 └─1372 /usr/sbin/nagios --worker /var/spool/nagios/cmd/nagios.qh
                   └─1373 /usr/sbin/nagios --worker /var/spool/nagios/cmd/nagios.qh
                     └─1374 /usr/sbin/nagios --worker /var/spool/nagios/cmd/nagios.qh
                       └─1375 /usr/sbin/nagios --worker /var/spool/nagios/cmd/nagios.qh
                         └─1376 /usr/sbin/nagios --worker /var/spool/nagios/cmd/nagios.qh
                           └─1377 /usr/sbin/nagios --worker /var/spool/nagios/cmd/nagios.qh
                             └─1378 /usr/sbin/nagios --worker /var/spool/nagios/cmd/nagios.qh
                               └─1386 /usr/sbin/nagios -d /etc/nagios/nagios.cfg

Mar 12 18:31:58 localhost.localdomain nagios[1364]: wproc: stderr line 02: s-nail: ... message not
Mar 12 18:31:58 localhost.localdomain nagios[1364]: wproc: stdout line 01: /var/spool/nagios/dead.lob
Mar 12 18:36:58 localhost.localdomain nagios[1364]: SERVICE NOTIFICATION: nagiosadmin;localhost;Curre
Mar 12 18:36:58 localhost.localdomain nagios[1364]: SERVICE ALERT: localhost;Current Load;WARNING;HAR
lines 1-26
```

Centos Prometheus status it failed maybe because of my configuration

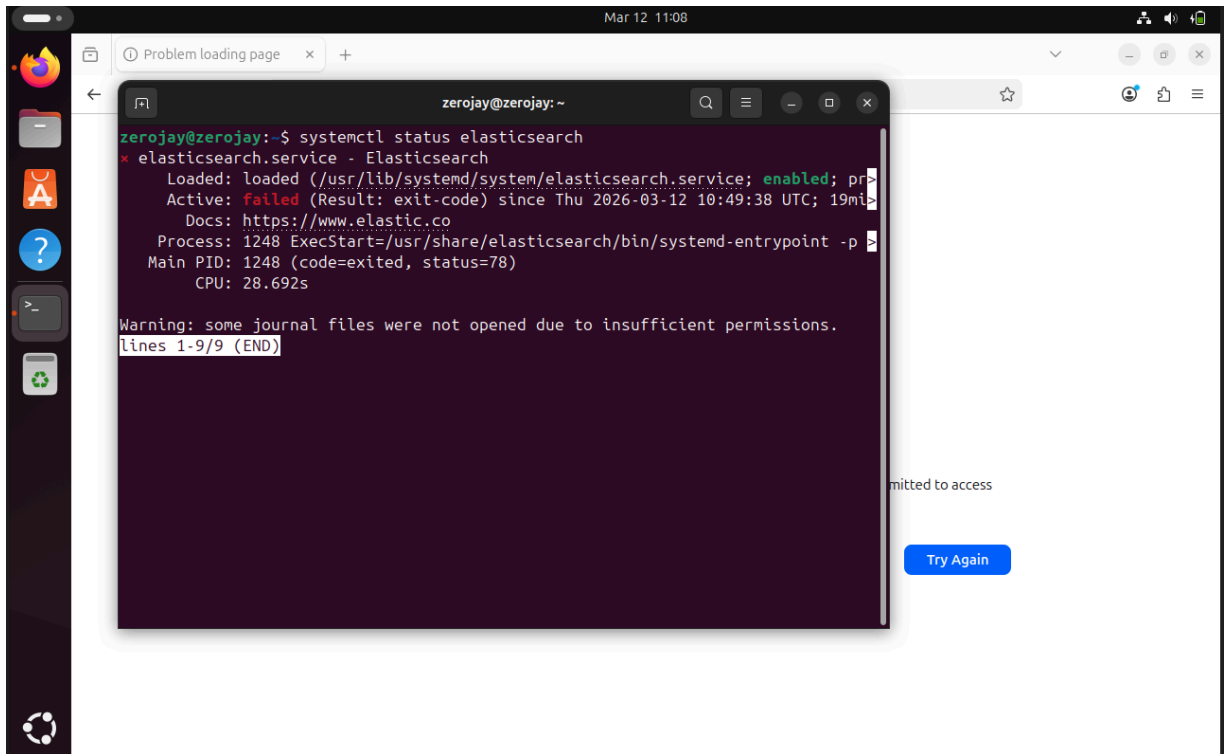
```
zerojay@localhost:~ - systemctl status prometheus

Mar 12 18:31:58 localhost.localdomain nagios[1364]: wproc: stderr line 02: s-nail: ... message not
Mar 12 18:31:58 localhost.localdomain nagios[1364]: wproc: stdout line 01: /var/spool/nagios/dead.lob
Mar 12 18:36:58 localhost.localdomain nagios[1364]: SERVICE NOTIFICATION: nagiosadmin;localhost;Curre
Mar 12 18:36:58 localhost.localdomain nagios[1364]: SERVICE ALERT: localhost;Current Load;WARNING;HAR

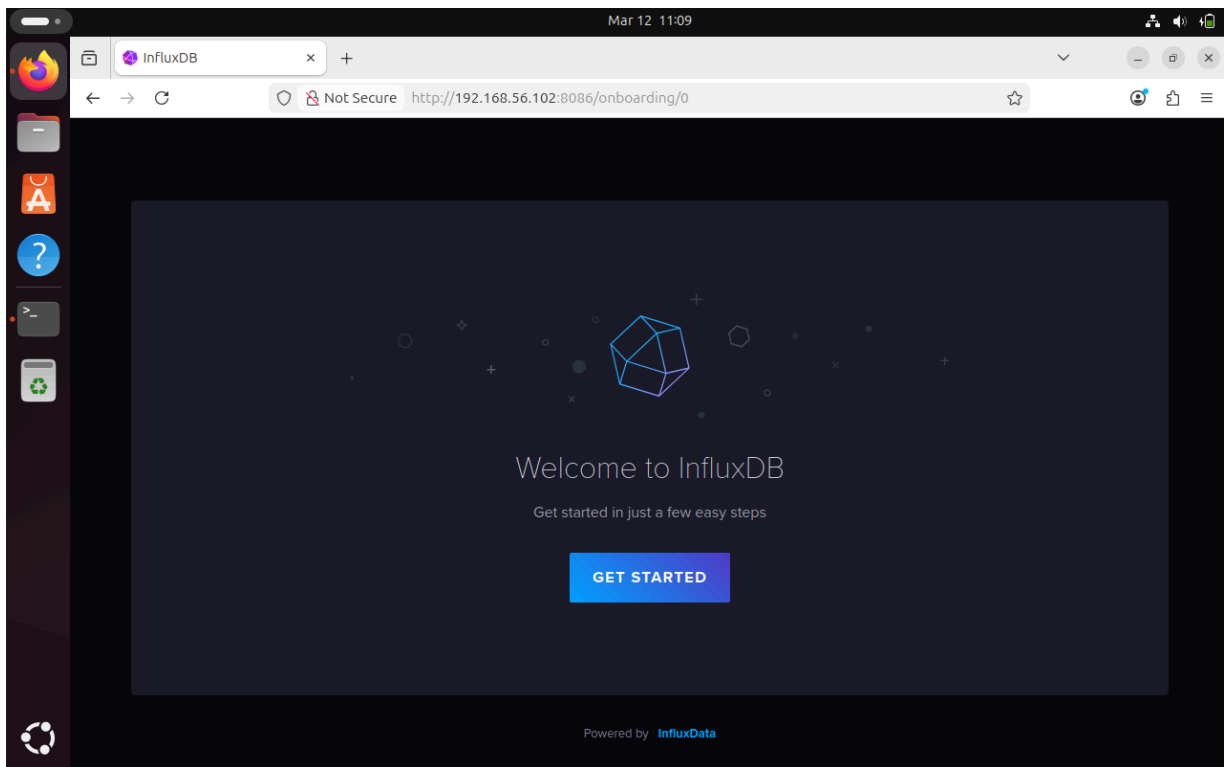
[2]+  Stopped                  systemctl status nagios
zerojay@localhost:~$ systemctl status prometheus
* prometheus.service - Prometheus
   Loaded: loaded (/etc/systemd/system/prometheus.service; enabled; preset: disabled)
   Active: failed (Result: exit-code) since Thu 2026-03-12 18:58:31 PST; 8min ago
   Duration: 1.023s
   Invocation: 74691bce3af9409faa6cd4270ff39a76
   Process: 34609 ExecStart=/usr/local/bin/prometheus --config.file=/etc/prometheus/prometheus.yml
   Main PID: 34609 (code=exited, status=1/FAILURE)
   Mem peak: 16.3M
   CPU: 700ms

Mar 12 18:58:30 localhost.localdomain systemd[1]: Started prometheus.service - Prometheus.
Mar 12 18:58:31 localhost.localdomain prometheus[34609]: time=2026-03-12T10:58:31.065Z level=INFO sou
Mar 12 18:58:31 localhost.localdomain prometheus[34609]: time=2026-03-12T10:58:31.065Z level=INFO sou
Mar 12 18:58:31 localhost.localdomain prometheus[34609]: time=2026-03-12T10:58:31.066Z level=INFO sou
Mar 12 18:58:31 localhost.localdomain prometheus[34609]: time=2026-03-12T10:58:31.082Z level=INFO sou
Mar 12 18:58:31 localhost.localdomain prometheus[34609]: time=2026-03-12T10:58:31.101Z level=INFO sou
Mar 12 18:58:31 localhost.localdomain prometheus[34609]: time=2026-03-12T10:58:31.110Z level=ERROR sou
Mar 12 18:58:31 localhost.localdomain systemd[1]: prometheus.service: Main process exited, code=exite
Mar 12 18:58:31 localhost.localdomain systemd[1]: prometheus.service: Failed with result 'exit-code'.
lines 1-19/19 (END)
```

elastic search on node 102 failed because of configuration



Influx db working on port 8086 after successful configuration



Kibana also working on Node 103



Grafana working on node 103 in port : 3000



GitHub link:

<https://github.com/qjfetalino-netizen/Midtermexam.git>

Conclusions: (link your conclusion from the objective)

In conclusion, the midterm project successfully demonstrated how Ansible can automate the deployment of multiple services across different virtual machines using a role based structure. The setup covered Elasticsearch, InfluxDB, Apache with PHP, Kibana, Grafana, MariaDB, Logstash, Prometheus, and Nagios, while also showing the importance of proper inventory grouping, variable management, and service mapping. Several problems were encountered during the process, including Git authentication issues, permission problems on local files and service directories, missing templates and handlers, undefined variables not being read from the configuration file, dependency conflicts on the CentOS server, and service startup failures caused by configuration or access rights. These challenges were gradually resolved by correcting file ownership, adding defaults inside roles, fixing handler and template paths, simplifying package installation where needed, and validating services after deployment. Overall, the activity highlighted not only the automation process itself, but also the practical troubleshooting skills needed to make a complex multi service deployment work properly.